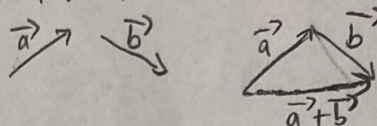


Total
17 / 17

1. Explain the "Triangle Law of Addition" between two vectors. Include a diagram in your explanation.

Add 2 vectors tip to tail, then draw a line connecting the tail of the first to the tip of the second, this is your resultant vector, ex:



(2)

3. Explain how you would determine the value(s) of k if $\vec{a} = k\vec{b}$ and $|\vec{a}| = 16|\vec{b}|$.

I see that they took the absolute value of both sides, then applied it to the 16, b/c 16 is the only thing multiplying $\vec{b}$, it must be the k value, however b/c the absolute value was applied, it can also be negative. $\therefore k = \pm 16$

(1)

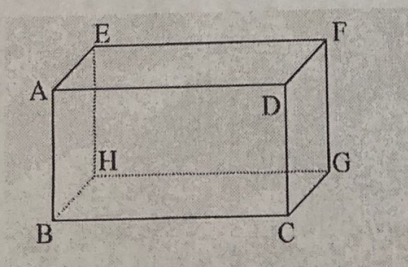
KNOWLEDGE:

1. Given $\vec{a} = 2\vec{i} + 4\vec{j} - 3\vec{k}$, $\vec{b} = -3\vec{j} + \vec{k}$, and $\vec{c} = 3\vec{i} - 2\vec{k}$, simplify $2\vec{a} + 4\vec{b} - \vec{c}$.

$$2\vec{a} + 4\vec{b} - \vec{c} = 4\vec{i} + 8\vec{j} - 6\vec{k} - 12\vec{j} + 4\vec{k} - 3\vec{i} + 2\vec{k} = \vec{i} - 4\vec{j}$$

(2)

2. Given the rectangular prism below, express each of the following as a single vector.



a) $\vec{CG} + \vec{CD} + \vec{CB} = \vec{CE}$

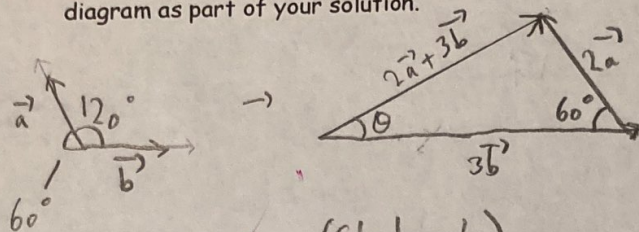
b) $\vec{HG} - \vec{CG} + \vec{BA} = \vec{HD}$

c) $\vec{FB} + \vec{AF} + \vec{CD} = \vec{0}$

$= \vec{FD} + \vec{DC} + \vec{CB} + \vec{AD} + \vec{DF} + \dots$

(3)

3. If $|\vec{a}| = 4$, $|\vec{b}| = 3$, and the angle between them is 120° , determine $2\vec{a} + 3\vec{b}$. Include a vector diagram as part of your solution.



using sine law (find angle)

$$\frac{\sin \theta}{8} = \frac{\sin 60^\circ}{\sqrt{73}}$$

$$\angle \theta = \sin^{-1} \left(\frac{8 \sin 60^\circ}{\sqrt{73}} \right) = 54.2^\circ$$

APPLICATION:

1. If $3\vec{x} - 2\vec{y} = \vec{a}$ and $5\vec{x} - 3\vec{y} = \vec{b}$, express the vectors $\vec{x}$ in terms of $\vec{a}$ and $\vec{b}$.

$$3\vec{x} = \vec{a} + 2\vec{y}$$

$$\vec{x} = \frac{\vec{a} + 2\vec{y}}{3}$$

$$5 \left(\frac{\vec{a} + 2\vec{y}}{3} \right) - 3\vec{y} = \vec{b}$$

$$\frac{5\vec{a} + 10\vec{y}}{3} - 3\vec{y} = \vec{b}$$

$$5\vec{a} + 10\vec{y} - 9\vec{y} = 3\vec{b}$$

$$\vec{y} = \frac{3\vec{b} - 5\vec{a}}{2}$$

$$\vec{x} = \frac{\vec{a} + 2(3\vec{b} - 5\vec{a})}{3}$$

$$\vec{x} = \frac{\vec{a} + 6\vec{b} - 10\vec{a}}{3} = \frac{6\vec{b} - 9\vec{a}}{3} = 2\vec{b} - 3\vec{a}$$

$$2m(\vec{x} + \vec{y}) + \frac{1}{2}n(\vec{x} + 4\vec{y} - \vec{z}) = -\frac{1}{2}\vec{x} - 5\vec{y} + \frac{3}{2}\vec{z}$$

2. Determine the values of m and n if possible:

(3) $\times 2$ everything $\rightarrow 4m(\vec{x} + \vec{y}) + n(\vec{x} + 4\vec{y} - \vec{z}) = -\vec{x} - 10\vec{y} + 3\vec{z}$

$$4m\vec{x} + 4m\vec{y} + n\vec{x} + 4n\vec{y} - n\vec{z} = -\vec{x} - 10\vec{y} + 3\vec{z}$$

$$4m\vec{x} + n\vec{x} = -\vec{x}$$

$$4m + n = -1$$

$$4m - 3 = -1$$

$$4m = 2$$

$$m = \frac{1}{2}$$

$$4m\vec{y} + 4n\vec{y} = -10\vec{y}$$

$$4m + 4n = -10$$

$$4m + 12 = -10$$

$$4m = -22$$

$$m = -\frac{11}{2}$$

$$-n\vec{z} = 3\vec{z}$$

$$-n = 3$$

$$n = -3$$

$$\therefore n = -3, m = \frac{1}{2}$$

check:

$$1(\vec{x} + \vec{y}) - \frac{3}{2}(\vec{x} + 4\vec{y} - \vec{z}) = \vec{x} + \vec{y} - \frac{3}{2}\vec{x} - 6\vec{y} + \frac{3}{2}\vec{z} = -\frac{1}{2}\vec{x} - 5\vec{y} + \frac{3}{2}\vec{z}$$

equal to original right side